UNIT 2 – EVOLUTION

ORIGIN OF LIFE

THEORIES OF EVOLUTION

- Lamarkism
- Darwinism

EVIDENCES OF ORGANIC EVOLUTION

- Morphological & Anatomical evidences
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What is Evolution?

Evolution mainly deals with the origin of life on earth. The conditions and the forms of life on earth were entirely different from what we see today. Everything evolved from one form to another for a better chance of survival.

THE ORIGIN OF THE UNIVERSE

The origin of life begins with the origin of the universe. The universe, an old vast and empty space comprising galaxies, originated around 20 billion years ago. There was nothing but blackness filled with gas and dust. The **Big Bang Theory** is the most accepted theory regarding the origin of the planet earth and the existence of different life forms on it. According to this theory, the universe is a result of a massive explosion which occurred 20 billion years ago. Whether it is a hypothesis or a fact, a new universe was formed. The atmospheric condition after the explosion became more stable. The temperature reduced and gasses like hydrogen and helium formed which led to the formation of galaxies of today.

Later, after 10 billion years, the earth was formed which was covered by water vapour, methane, carbon dioxide, and ammonia. There was no atmosphere but only gases and moisture. The powerful rays of the sun stimulated and hastened evolution. By making and breaking bonds between gas molecules, the earth came out with a new face. After millions of years i.e., once the earth's atmosphere was stable, the first life on earth came into existence (around 4 million years ago). There began the story of the origin of life on earth.

ORIGIN OF LIFE

As per the findings in evolutionary biology, the universe, earth and the variety of life forms are the results of evolution.

Here, let us know in brief about the origin of life on earth.

EARLY THEORIES

1) **THEORY OF SPECIAL CREATION**
   - This theory states that living organisms on the earth were created by **Divine Power**. The greatest supporter of this theory was father Suarez. According to the Bible, life and everything was created by God in 6 days. The first man, Adam and the first woman, Eve were created by God.
   - According to Hindu mythology, the world was created by God Brahma. The first man was Manu and the first woman was Shradha. Special creation theory lacks scientific evidence so it is not accepted.

2) **COSMOZOIC THEORY or PANSPERMIA**
   - This theory is proposed by **Richter** and supported by **Arrhenius.**
   - Life is distributed all over the cosmos in the form of resistant spores.
   - Resistant spores of living organisms, called cosmozoa, together with cosmic dust might have reached the earth accidentally from other planets of the universe.

3) **THEORY OF CATASTROPHISM**

   According to this theory earth was subjected to periodic catastrophes. These catastrophes destroyed the life from time to time and created new and special form of life after each destruction

4) **THEORY OF SPONTANEOUS GENERATION (ABIOGENESIS)**

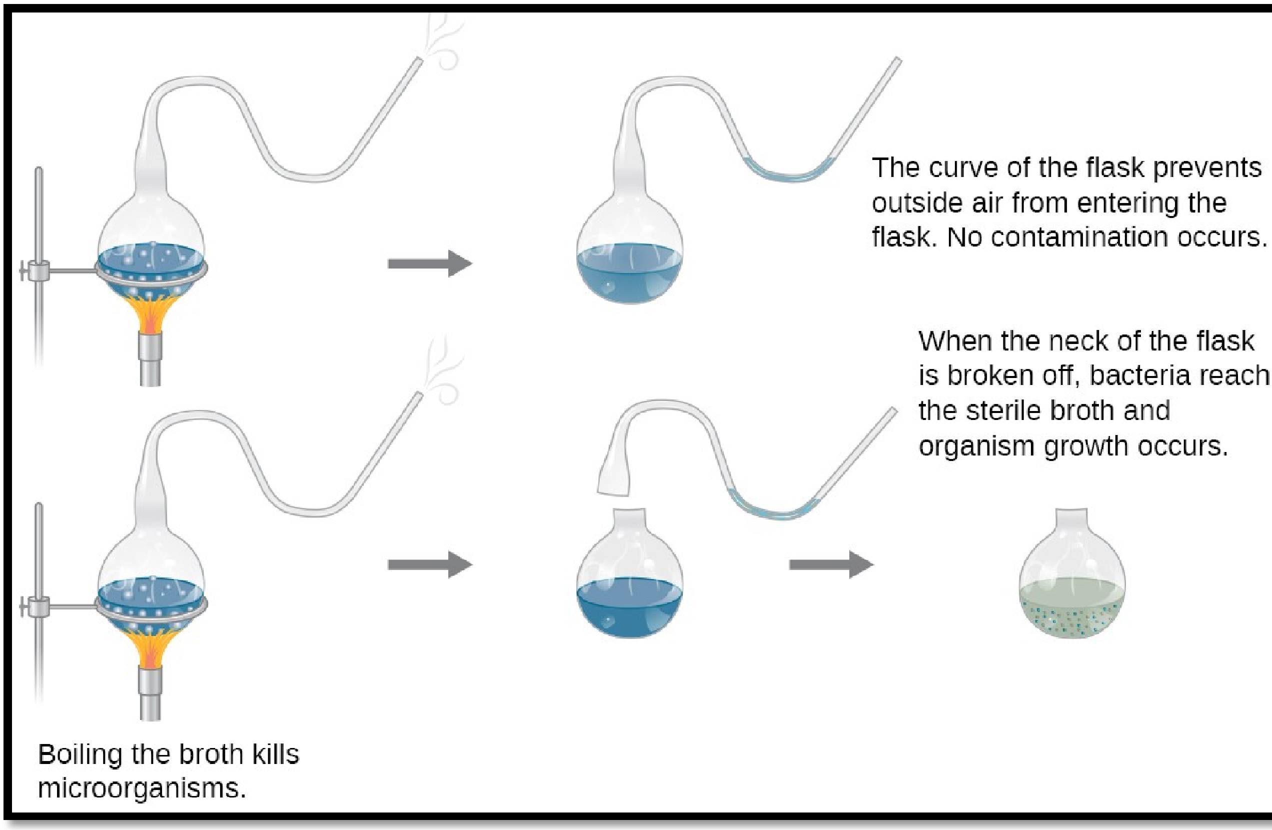
   This theory states that life originated from nonliving things in a spontaneous manner. In ancient Egypt, it was believed that the mud of the Nile could give rise to frogs, toads, snakes, mice and even crocodiles when warmed by the sun. Francesco Redi, Spallanzani and Louis Pasteur experimentally disproved abiogenesis theory.

5) THEORY OF BIOGENESIS

This theory states that living organisms originate from pre-existing organisms. Louis Pasteur confirmed the theory of biogenesis by his swan-neck flask experiment.

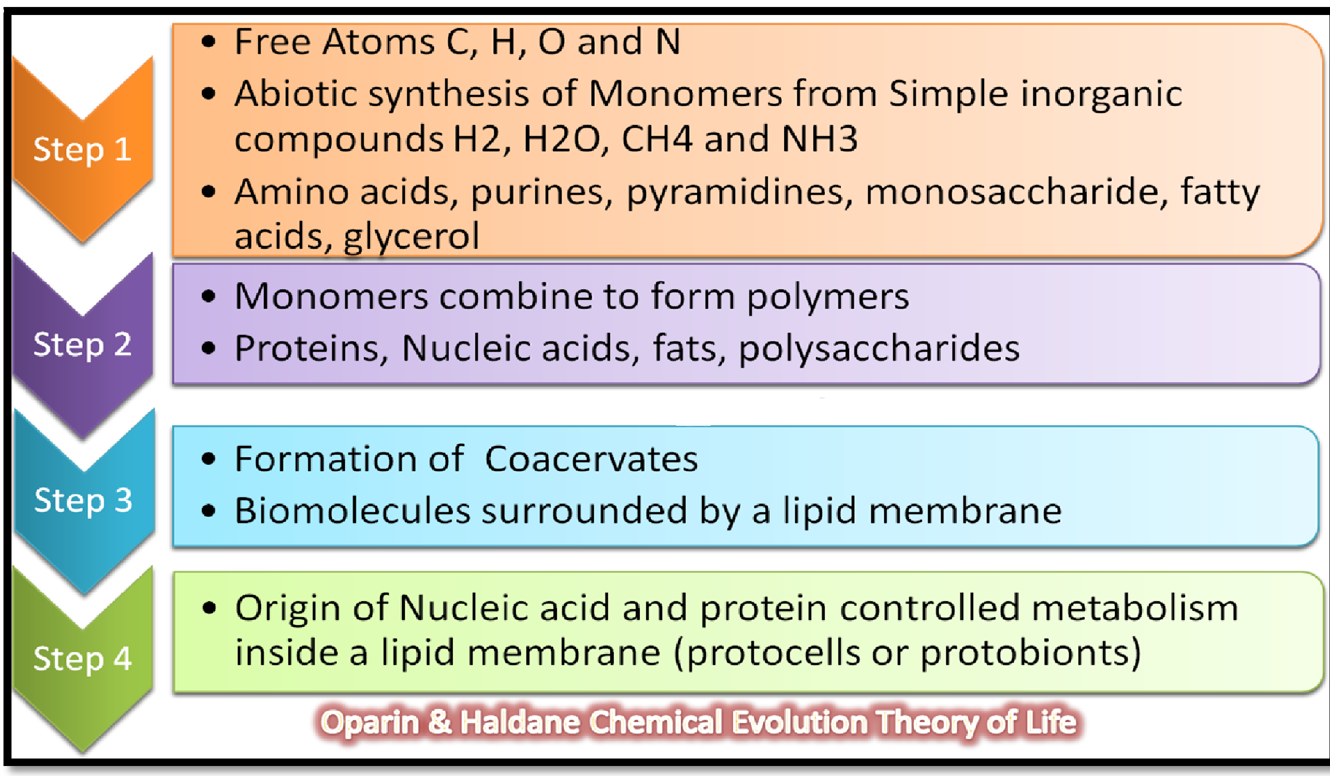
Pasteur's Experiment

Louis Pasteur, a French scientist took broths in a long necked flask and then he bent the neck of the flask. He boiled the broths in the flask to kill any microorganisms that might be present in them. The curved neck acted as a filter. If the flask with 'swan neck' (curved neck) is kept for months together, no life appeared, as the germ laden dust particles in the air were trapped by the curved neck which serves as filter. If the swan neck was broken off, the broths developed colonies of moulds and bacteria. Thus, he showed that the source of the microorganisms for fermentation or putrefaction such as for milk, sugar and wine, etc., was the air and the organisms did not arise from the nutrient media.



6) **Theory of organic evolution**

The origin of primordial life on the earth was associated with the origin of universe. Primitive organisms evolved spontaneously from the inorganic mater as a result of formative action of physical forces like electric discharge of lightening, U.V radiations and radiations of radioactive elements. This hypothesis was strongly supported by Haldane, Oparin, Urey and Miller. Oparin described the origin of life in his book "The origin of life on the earth". According to Haldane and Oparin the first phase of the origin of life was the spontaneous generation of early molecules. These molecules later transformed into protobionts (molecules with limited metabolic activity). Finally the protobionts evolved into early living organisms. Thus origin of life is a phenomenon of chemical evolution that led to biological evolution.



THEORIES OF EVOLUTION

Evolution simply means "change". According to Darwin "Evolution is Descent with modification". Evolution is defined as a gradual and orderly change from one condition to other. Such a kind of change in the living world is called organic evolution. The gradual development and emergence of complex organisms from simple organisms over a period of time is called organic evolution. Evolution is a naturally occurring process. It is a slow, continuous and irreversible process of change. Several theories have been put forward to explain the phenomenon of evolution. Most important ones are

1) **LAMARCKISM**
2) **DARWINISM**

LAMARCKISM

Jean Baptiste Lamarck for the first time suggested a complete theory of evolution which is known as "**Theory of Inheritance of acquired characters/Theory of use and disuse**" which is also known as Lamarckism. He published his ideas of evolution in his book "***Philosophie Zoologique***" in 1809.

Salient features of Lamarckism

1) All living things and their component parts are continuously increased due to internal **vital forces**.

2) Environment keeps on changing and thus influences the organisms by creating new needs. The new needs lead to the development of new structures.

3) When an organ is put into greater and constant use, that particular organ develops well, at the same time when an organ is not used for a long time, it decreases and in

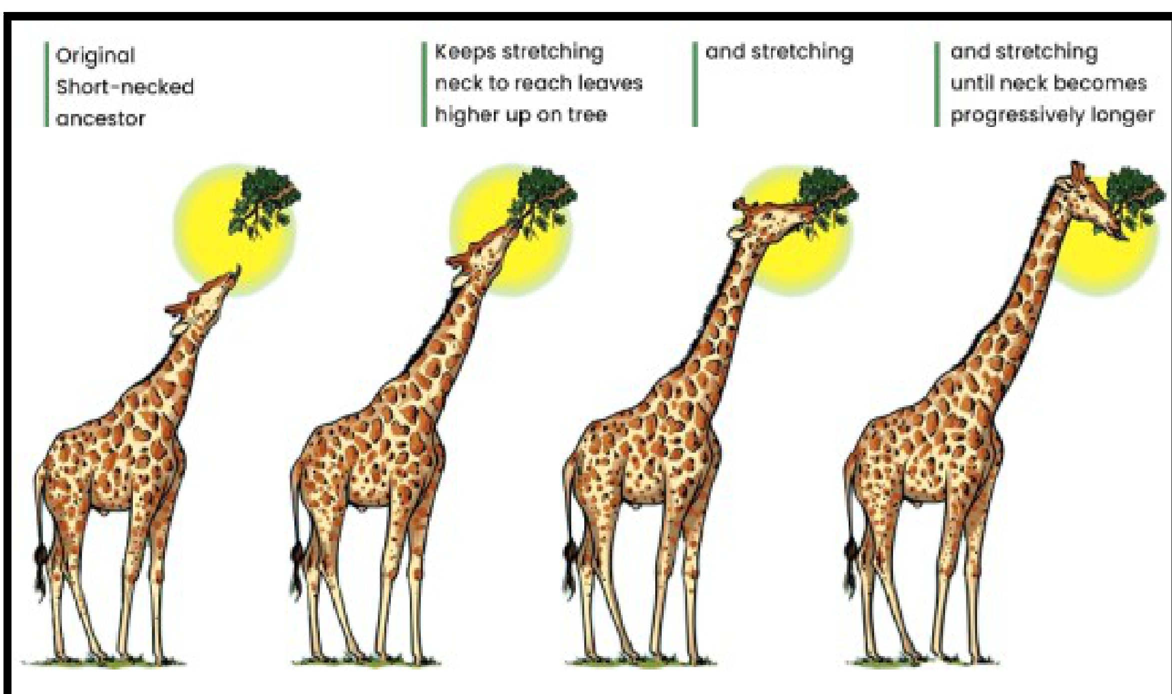
due course it degenerates and disappears completely from the organism. This is called “Theory of Use and Disuse”.

4) The characters developed by the animals during their life time in response to the environmental changes are called acquired characters. These are transmitted to the offspring. This is called inheritance of acquired characters.

5) As the process is continued, after several generations a new species will be formed.

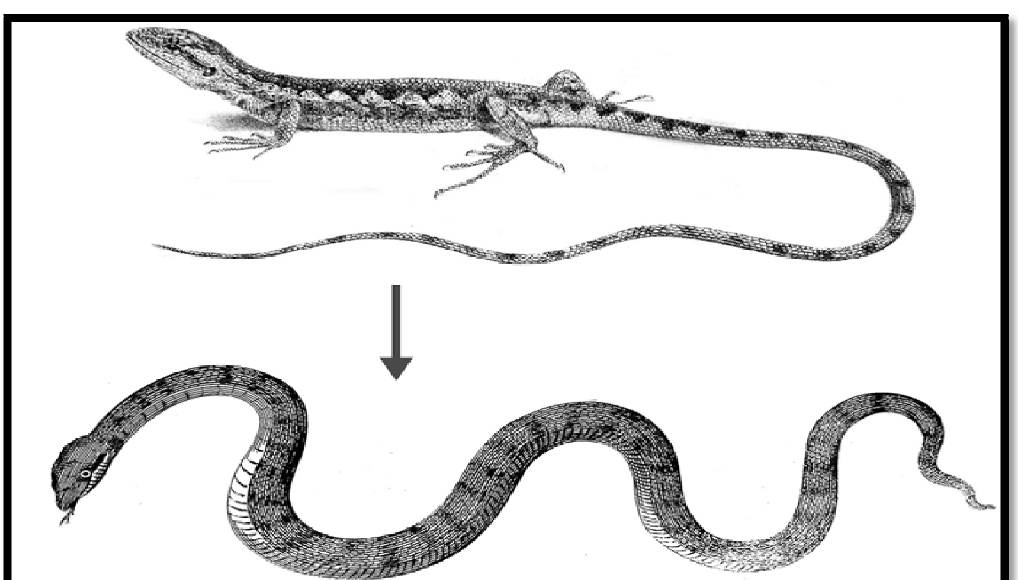
EXAMPLES

Long necked giraffe -- The evolution of long necked giraffe took place from short-necked giraffe due to continuous stretching of the neck muscles in order to find food from tall trees. In the beginning, the short-necked giraffe used to eat the grasses. Later on, sources of grass on land reduced and it forced to eat the leaves of tall trees stretching of neck is continuous and is gradually transmitted to offspring’s.



Aquatic Birds with Webbed Toes - The evolution of Ducks occurred from terrestrial ancestors. Due to the lack of food on land, these aquatic animals need to go to water. Hence, some web-like structures are developed between the toes in them making their lives easy in the water.

Loss of limbs of snake: The continuous creeping through holes and crevices made snakes body elongated and due to continuous disuse of limbs because they hinder while creeping in burrows results in loss of their limbs.

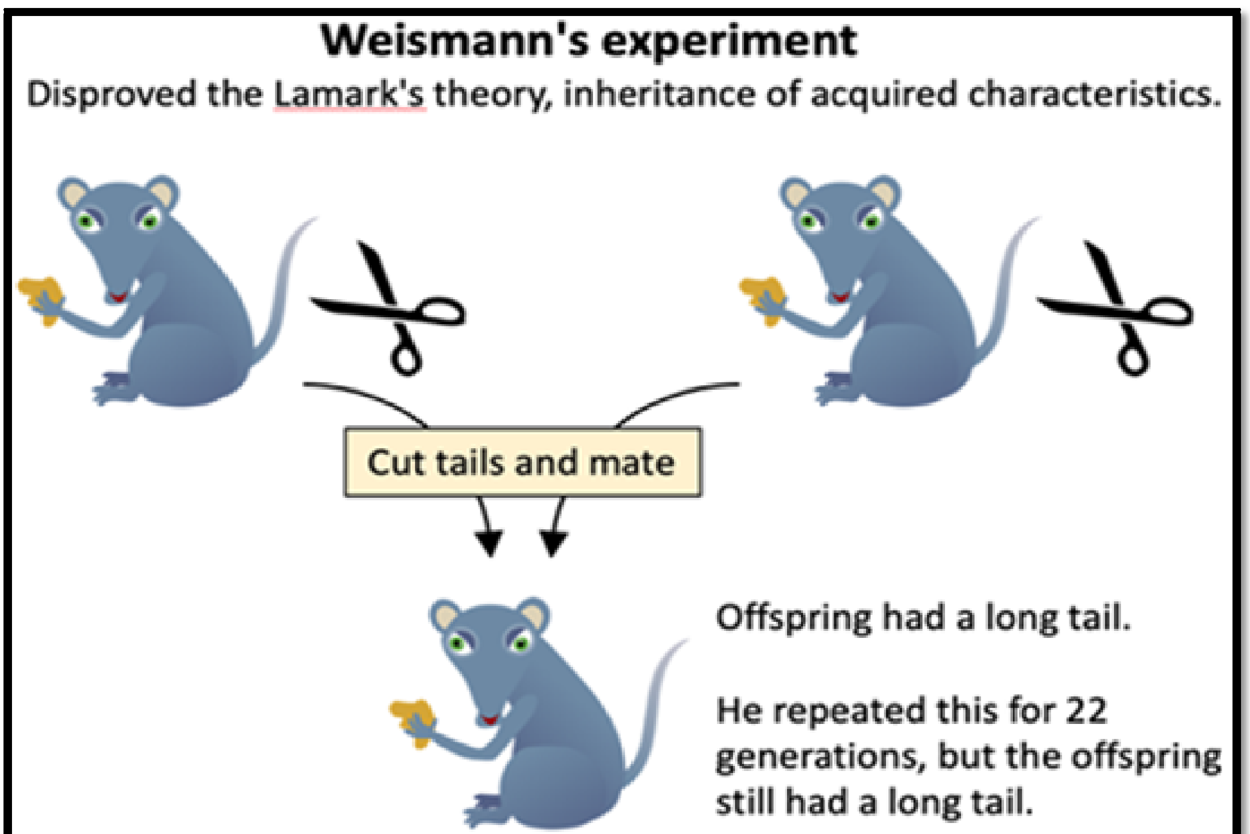


CRITICISM

However, Lamarck's Theory was subject to severe criticism.

Weismann proposed the “**Theory Of Continuity Of Germplasm**” where he said the germplasm (protoplasm of germ cells) is inherited and the somatoplasm is not transmitted to the next generation.

- For example the muscle of a wrestler doesn’t get transmitted to his kids because it is an acquired character by him which causes changes in his somatic cells only.
- Weismann proved this theory by performing an experiment in which he cut off the tail of mice for 22 generations and allowed them to breed. Every offspring was born with a tail, never once a tailless mouse was born. Because cutting off the tail was a trait that the mice acquired during its lifetime and has nothing to do with the germ cells of the mice. Therefore the trait was not inherited to the next generation.



NEO-LAMARCKISM

Modified form of Lamarckism is called NEO-LAMARCKISM

It proposes that:

1) Environment does influence an organism and change its heredity.
2) Atleast some of the variations acquired by an individual can be passed on to the offspring.
3) Internal vital force do not play any role in evolution
4) Only those variations are passed on to the offspring which affect germ cells or where somatic cells give rise to germ cells.

DARWINISM

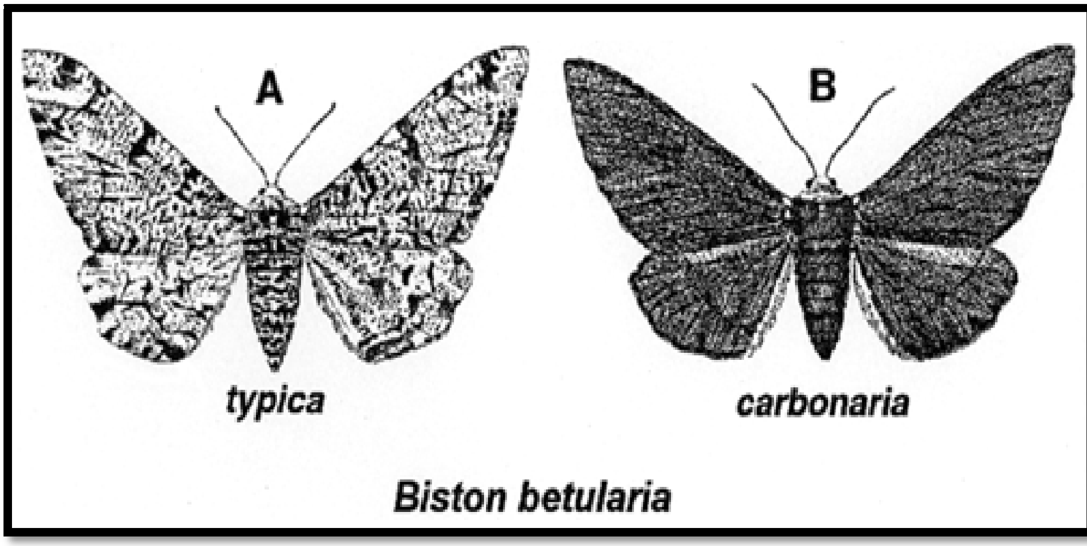
Charles Robert Darwin (1809-1882) was an English naturalist born on 12TH February, 1809 at Shrewsbury, England. He was appointed as a Naturalist in 1831 upon a world survey-ship of British government **H.M.S. Beagle**. He went on voyage for five years (1831 to 1836) and explored the fauna and flora of a number of continents and islands (**Galapagos**, Ecuador, Peru, Chili, Argentina). Darwin presented the summary of his theory in a joint paper entitled, "***Origin of species***" in 1858. He published his findings in the book entitled "**The origin of species by natural selection**" or "**The preservation of favoured races in the struggle for life**". Charles Darwin is renowned as the "Father of Evolution" because of his role in the development of evolutionary.

Darwinism or natural selection theory proposed by Charles Darwin, explains the mechanism of evolution. The mechanism is clearly stated in Darwin's book "Origin of Species". The basic postulates of Darwinism are:

1) **Over production**: Power of multiplication or reproduction is one of the prime features of life. Organisms reproduce at a very high rate. For example A female salmon fish produces 28,000,000 eggs in a season, *Ascaris lumbricoides* passes 7 lakhs eggs in 24-hours, an oyster liberates 114,000,000 eggs at a single spawning and *Rana catesbiana* produces 20,000 eggs annually.
2) **Limited natural resources:** However, an abnormal increase in population is not observed in nature. The population of each species remains more or less constant because the offspring die in large numbers before they become reproductively active. The food and other sources do not increase in the same rate of population increase. Hence, more number of individuals are eliminated as available resources support a limited number of organisms. A great competition exists wherein the individual tries to become better than the other to protect itself from the danger of being eliminated.

3) **Struggle for existence:** Over population results in severe competition. Darwin called in struggle for existence. The struggle is threefold.
   a) *Intraspecific struggle* – the struggle found among the individuals of a species
   b) *Interspecific struggle* – found among the animals of different species.
   c) *Struggle with the environment / inanimate nature*- living organism struggle with the adverse environmental conditions like floods, cold waves, heat waves, earthquakes etc.
4) **Variations and Heredity**: During the competition, every individual tries to become better and shows variation. But all the variations are not significant from the evolutionary point of view, some may be useful and some harmful. The useful variations are inherited in the progeny of those organisms in which they arose and therefore, the progeny have better chances of survival.
5) **Survival of the fittest or natural selection**: During the struggle for existence only those individuals who are better adapted to the environment and show favourable variations are selected. The less fit and unfit individuals are rejected by nature. Darwin stated that the variations are sorted out naturally.

A classical example of natural selection in the wild is the case of peppered moth *Biston betularia*. These occur in two phenotypes, grey and black. Prior to industrial revolution the gray moths succeeded to camouflage the light trunks of the trees. With the industrial revolution more soot was released due to the burning of coal. Tree barks became black. Grey moths were easily identified and were more predated by the birds as a result they decreased in number and dark moths increased in the population. Therefore natural selection favored the melanic moths to reproduce more successfully. Natural selection of darker forms in response to industrial pollution is known as industrial melanism.



6) **Origin of species**: According to Darwin, new variations appear in every generation and get inherited to the new generation. As a result, the offsprings are distinct from their ancestors. This results in the formation of new species. Gradual modification of the existing species leading to the formation of new species is known as speciation.

Criticism

a) Darwin was unable to explain the mechanism of inheritance of characters. Darwin proposed the theory of pangenesis to explain this phenomenon. He said that every cell or organ produces minute hereditary particles called pangenes or gemmules. These were carried through the blood and deposited in the gametes. This theory was not accepted.
b) Darwin was unable to explain the source of variations in organisms.
c) According to natural selection, only useful organs are favoured by natural selection. The existence of vestigial organs in organisms could not be explained.
d) In some species of deer, the antlers develop beyond the stage of usefulness. These structures are of no functional significance to the animal.

EVIDENCES OF ORGANIC EVOLUTION

Organic evolution is the progressive development of simple organisms to the complex organisms over a period of time. The evidences of organic evolution includes

1) Morphological and anatomical evidences
2) Embryological evidences
3) Paleontological evidences

1) MORPHOLOGICAL AND ANATOMICAL EVIDENCES

Morphology refers to the external structures. Anatomy refers to the internal structures. These structures provide ample evidences for evolution. The morphological and anatomical evidences include

a. Homologous organs
b. Analogous organs
c. Vestigial organs
d. Connecting links

Homologous organs (Divergent evolution)

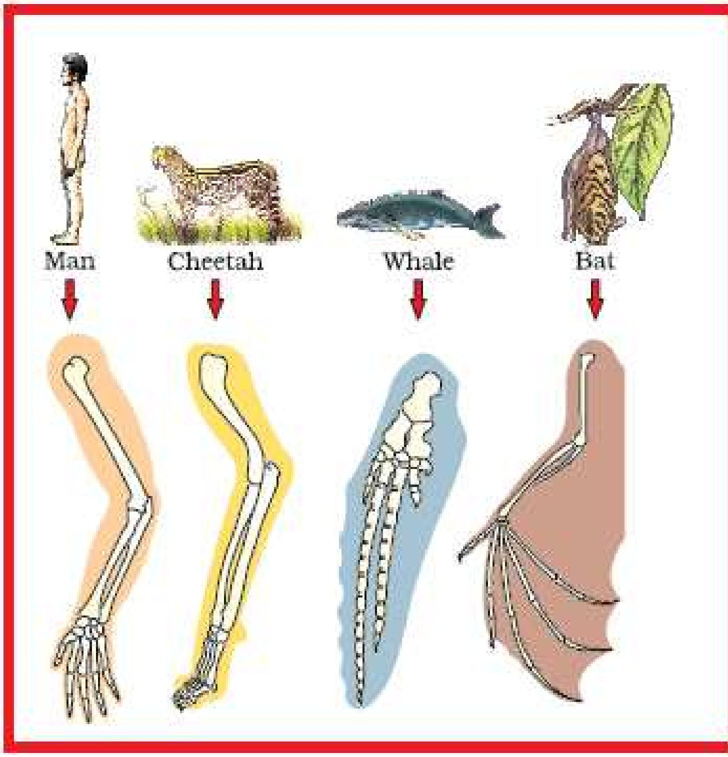
Organs which are fundamentally similar in origin and basic structure but perhaps modified for widely different functions, can only be explained through a common ancestry. These types of organs are called homologous organs.

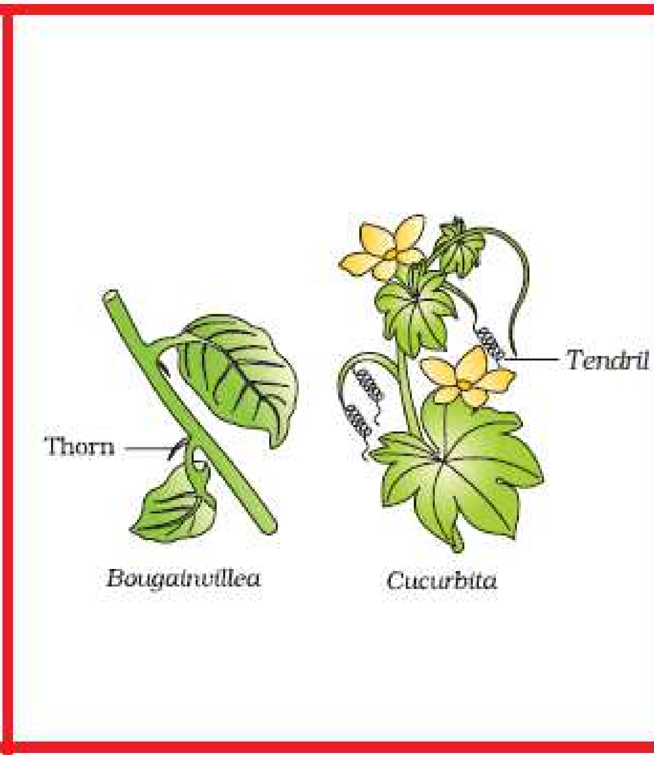
For example

- ✓ **Human, Cheetah, whales and bats (all mammals) share similarities in the pattern of bones of forelimbs**. Though these forelimbs perform different functions in these animals, they have similar anatomical structure – all of them have humerus, radius, ulna, carpals, metacarpals and phalanges in their forelimbs.

Hence, in these animals, the same structure developed along different directions due to adaptations to different needs. Homology indicates common ancestry.

- ✓ In plants also, **the thorns of *Bougainvillea* and tendrils of *Cucurbita*** represent homology. Both of them are modified axillary buds. The thorn serves as a defense against grazing animals whereas the tendrils aid in providing support when climbing.





ANALOGOUS ORGANS (Convergent evolution)

The organs which are different in their basic structure but similar in form and function are called analogous organs.

For example

- ✓ **Wings of birds and butterfly**- They differ in origin and structure but look alike and perform the same function of flying. This similarity is achieved due to their existence under similar environmental conditions.

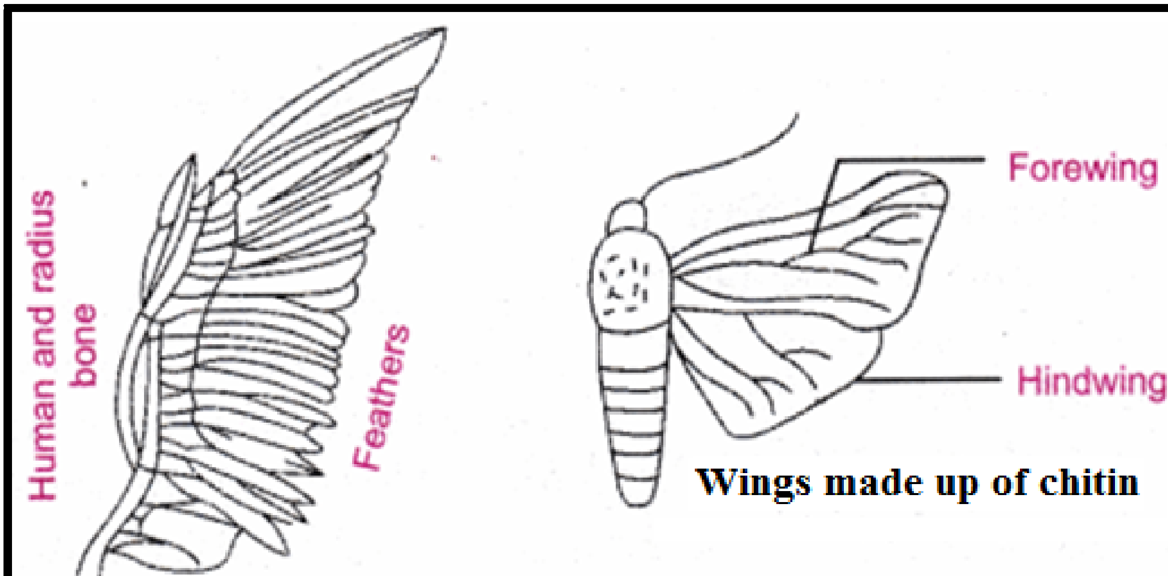
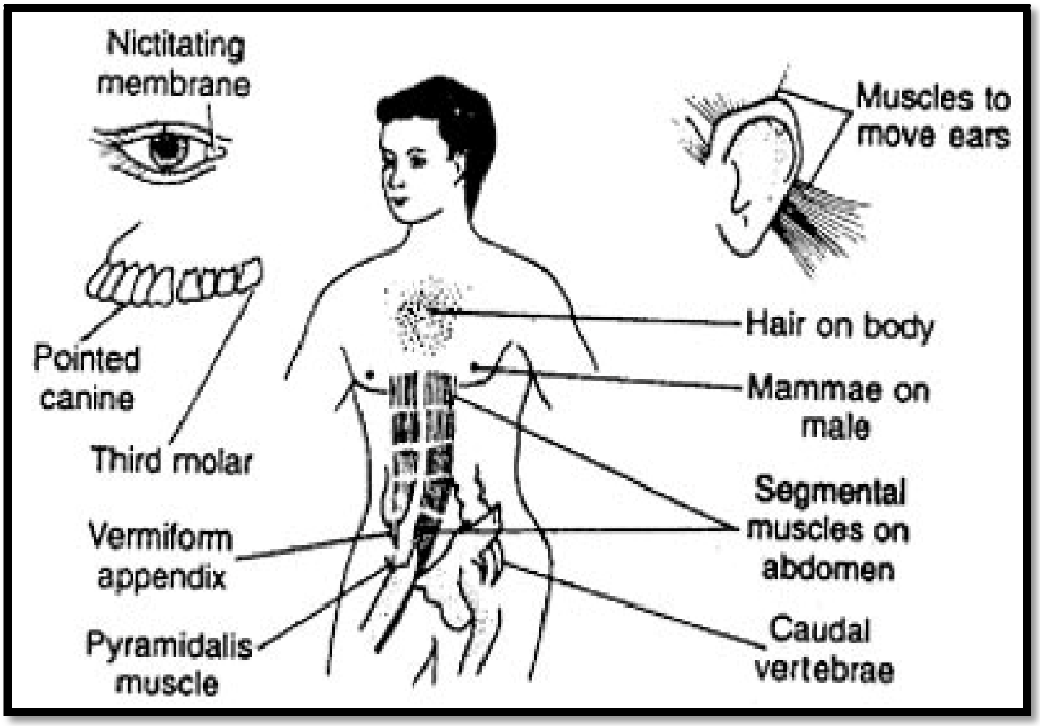


Fig. Wings of Bird and Butterfly

Vestigial organs

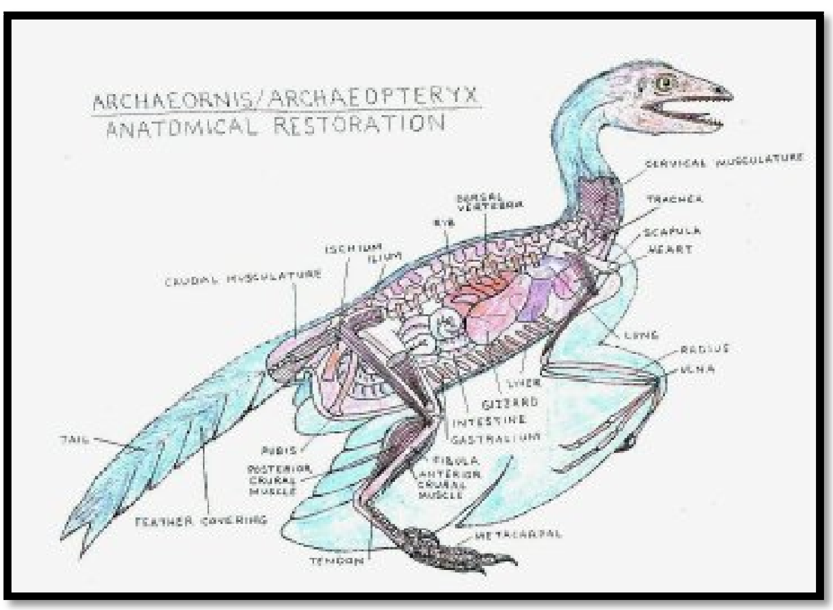
The organs which are functional in the ancestors and reduced and remained functionless in the present generations are called as vestigial organs. Examples of vestigial organs in man are Vermiform appendix, Muscles of the ear pinna etc.



Connecting links

The animals exhibiting characters of two adjacent groups are called connecting links.

Examples *Archaeopteryx* and *Archaeornis* were the first fossil birds which are quite different from the modern birds. They show few reptilian characters such as long tail, Jawed teeth, solid bones and few avian characters such as feathers, three digits in fore limbs etc. It shows that modern birds evolved from reptiles.



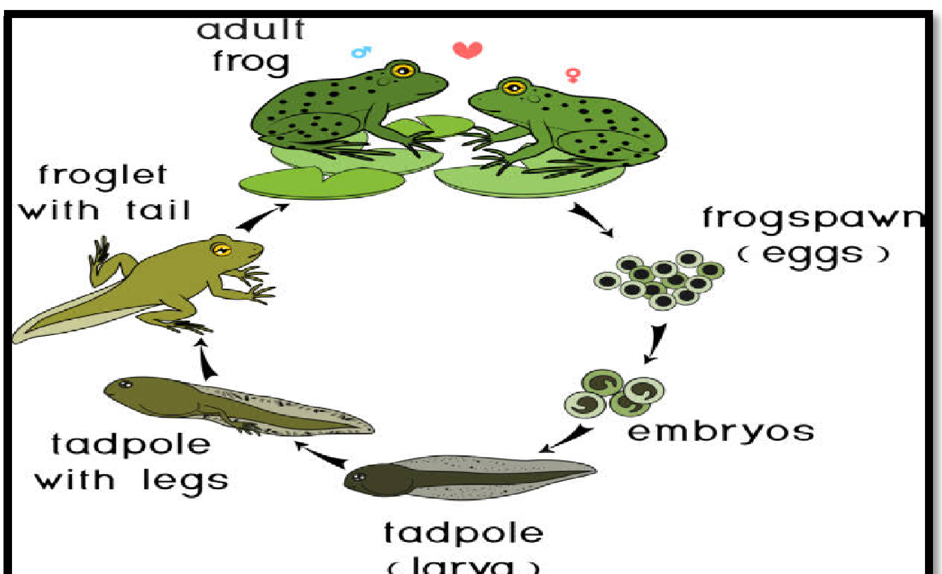
EMBRYOLOGICAL EVIDENCES

Embryology deals with the **development of an organisms form the egg to adult**. It includes a series of stages of development. A comparative study of the developmental stages, organs and organ systems give evidences for organic evolution.

Similarly, the **embryo of different vertebrates like fishes, amphibians, reptiles, birds and mammals are similar in their early stages and cannot be differentiated from one another**. As the development progresses they can be differentiated due to the development of specialized characters of their adults. This kind of similarity of the early embryos indicates that they have evolved from a common ancestor.

On the basis of the developmental history of the animals, **Haeckel proposed "Recapitulation theory or Biogenetic law"** which states that **"Ontogeny (embryonic development of an organism) recapitulates Phylogeny (Evolutionary history of the race or phylum)"**. This means that embryos, in their development, repeat the evolutionary history of their ancestors.

Example- Tadpole larva of frog looks like a fish, it has gills, fins, lateral line sense organs etc as in fish. It undergoes metamorphosis to become adult frog. It shows that the frog recapitulates its immediate ancestor, fish through tadpole larva which indicates that frog was evolved from fish.



PALEONTOLOGICAL EVIDENCES

Paleontology deals with the study of fossils. Fossils are defined as the remnants of the previously existing animals and plants preserved in the earth crust. Hard parts of the organisms (plants, animals etc) like bones, teeth, shells, wood etc are preserved in the rocks. By studying the fossils in different sedimentary layers of earth's crust, we can estimate the geological period in which they existed. The study also showed that life forms varied over time and certain life forms are restricted to certain geological life spans. Hence, new forms of life have arisen at different times in the history of earth. All this is called paleontological evidence.

For example, **Dinosaurs** got extinct from earth and their skeletons are being found from different areas of earth. Even some present day organisms resemble the extinct dinosaurs in structure which indicates that these modern organisms have evolved from dinosaurs.

1. Which theory attempts to explain to us the origin of universe?
    a) Explosion Theory
    b) Big-Bang Theory
    c) Black hole theory
    d) Gamma Theory
2. Who demonstrated that life originated from pre-existing cells
    a) Louis Pasteur
    b) Aristotle
    c) Morgan
    d) Stanley Miller
3. During origin of life, which among the following was not found in free form
    a) Hydrogen
    b) Oxygen
    c) Nitrogen
    d) Carbon
4. Miller in his experiment, synthesized simple amino acid from
    a) Methane, ammonia, oxygen, Nitrogen
    b) Hydrogen, methane, ammonia, water
    c) Ammonia, methane, carbon dioxide, water
    d) Hydrogen, water, oxygen, helium
5. According to spontaneous generation life originated from
    a) Micro organisms
    b) Living organisms
    c) From non living organisms
    d) From air
6. The universe is almost --------- years old
    a) 20 billion years
    b) 10 billion years
    c) 500 million years
    d) 300 million years
7. In Lamarck's view the key of organic evolution is that each progeny
    a) Shows struggle for existence
    b) Characters acquired by parental generation are inherited
    c) is similar to its parents
    d) Phylogeny is repeated in its ontogeny

8. The idea of use and disuse of organs was given by
    - **a) Lamarck**
    - b) Darwin
    - c) T H Morgan
    - d) Hugo De Vries

9. Embryological support for evolution was proposed by
    - a) Von Bayer
    - **b) Haeckel**
    - c) Darwin
    - d) T H Morgan

10. Who stated this theory – organ in use will develop and if not used will weaken to turn vestigial
    - **a) Lamarck**
    - b) Darwin
    - c) Morgan
    - d) De Vries

11. Germplasm theory against Lamarck's principle was put forward by
    - a) Darwin
    - b) Lamarck
    - c) De Vries
    - **d) Weismann**

12. This is an example of industrial melanism
    - a) Mutation
    - **b) Natural Selection**
    - c) Neo – Lamarckism
    - d) Overproduction

13. What is the scientific name of moth that demonstrated industrial melanism?
    - a) *Pisum sativum*
    - b) *Biston mothis*
    - **c) *Biston betularia***
    - d) *Periplaneta americana*

14. The force that initiates evolution is ______
    - **a) Variation**
    - b) Mutation
    - c) Extinction
    - d) Adaptation

15. Observation of species on ___________ heavily inspired Darwin's theory of evolution.
    a) Ilha da Queimada Grande
    b) Guatemala
    c) Faroe Islands
    d) Galapagos Islands

16. "**The Origin of Species**" was written by ______
    a) Lamarck
    b) De Vries
    c) Charles Darwin
    d) Francesco Redi

17. In some animals, the same structures develop along different directions due to adaptations to different needs, this is called as _____
    a) **Divergent Evolution**
    b) Convergent Evolution
    c) Parallel Evolution
    d) Saltation

18. Which among the following are examples of homologous organs?

    a) Wing of an insect and wing of a bird
    b) The arm of a human and **forelimb of cheetah**
    c) Leg of a dog and leg of a spider
    d) All the above

19. _____________ was considered as a connecting link between reptiles and birds.

    a) Pteranodon
    b) Avimimus
    c) Archaeopteryx
    d) Caudipteryx

20. Which of the following is the correct sequence of events in the origin of life?

I. Formation of protobionts.

II. Synthesis of organic monomers.

III. Synthesis of organic polymers.

IV. Formation of DNA-based genetic systems.

a) I,II,III,IV
b) I,III,II,IV
c) II,III,I,IV
d) II,III,IV,I